# 机械臂通信协议说明

## 一、 基本说明

1. client 为调用系统，server 为机械臂系统。双方通过 TCP 方式通信
2. server 的 IP 为 192.168.10.95，端口为 30485。
3. 协议数据每一包均用“json 数据域 + 帧尾”的形式发送

| 名称 | 内容 | 长度(byte) | 描述 |
|---|---|---|---|
| 数据域 | | | json 序列化数据内容 |
| 帧尾 | | | # |

## 二、 帧内容说明

### 1. MoveJ：关节运动

Client

```
{
    "command":"movej",
    "joint":[0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.8,0.9,1.0,1.1,1.2,1.3,1.4,1.5,1.6,1.7],
    "v":50
}#
```

Server

```
{
    "command":  "movej",
    "status":  "program_run_finish"
}#
```

说明：

- **joint：** 1-17 关节角度，单位：弧度，顺序为：腰/L1-L7/R1-R7/摇头/点头 ，-100 表示该关节不移动 。
- **v：**速度百分比例系数，0~100

- **status** ： program_run_finish/program_run_failed

### 2. MoveJ_Canfd：角度透传

Client

```
{
    "command":"movej_canfd",
    "joint":[0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.8,0.9,1.0,1.1,1.2,1.3,1.4,1.5,1.6,1.7],
    "freq": 50
}#
```

Server

```
{
    "command": "movej_canfd",
    "status": "program_run_finish"
}#
```

说明：

- **joint**：1-17 关节角度，单位：弧度，顺序为：腰/L1-L7/R1-R7/摇头/点头 ，-100 表示该关节不移动 。
- **status** ： program_run_finish/program_run_failed

### 3. MoveJ_P：关节空间规划到目标位姿

Client

```
{
    "command":"movej_p",
    "pose":[x, y, z, qw, qx, qy, qz, x, y, z, qw, qx, qy, qz],
    "v":50
}#
```

Server

```
{
    "command": "movej_p",
```

"status": "program_run_finish"

}#

说明：

- pose：left TCP (x, y, z, qw, qx, qy, qz) + right TCP (x, y, z, qw, qx, qy, qz)，单位：弧度, -100 表示该臂不移动
- **v**：速度百分比例系数，0~100
- **status** ： program_run_finish/program_run_failed

### 4. MoveL：直线运动

Client

{

"command":"movel",

"pose":[x, y, z, qw, qx, qy, qz, x, y, z, qw, qx, qy, qz],"

v":50

}#

Server

{

"command": "movel",

"status": "program_run_finish"

}#

说明：

- pose：left TCP (x, y, z, qw, qx, qy, qz) + right TCP (x, y, z, qw, qx, qy, qz)，单位：弧度, -100 表示该臂不移动
- **v**：速度百分比例系数，0~100
- **status** ： program_run_finish/program_run_failed

### 5. get_joint_position: 获取关节位置

Client

{

"command":"get_joint_position"

}#

Server

{

"command": "get_joint_position",

"joint":[0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.8,0.9,1.0,1.1,1.2,1.3,1.4,1.5,1.6,1.7],

"status": "program_run_finish"

}#

说明：

- **joint**：1-17 关节角度，单位：弧度，顺序为：腰/L1-L7/R1-R7/摇头/点头
- **status** ： program_run_finish/program_run_failed

### 6. get_tcp_pose: 获取 TCP 位姿

Client

{

"command":"get_tcp_pose",

}#

Server

{

"command": "get_tcp_pose",

"pose":[x, y, z, qw, qx, qy, qz, x, y, z, qw, qx, qy, qz],"

"status": "program_run_finish"

}#

说明：

- **pose**：left TCP (x, y, z, qw, qx, qy, qz) + right TCP (x, y, z, qw, qx, qy, qz)，单位：弧度
- **status** ： program_run_finish/program_run_failed

### 7. get_device_state: 获取设备状态

Client

{

"command":"get_device_state",

}#

Server

{

"command": "get_device_state",

"erroeCode": 0,

"hardwareErrors":[],

"softwareErrors":[],

"status": "program_run_finish"

}#

说明：

- erroeCode：0x00: 正常， 0x01:硬件错误，0x02：软件错误
- hardwareErrors: 如 0x01010001:表示硬件第一个电机掉线
- softwareErrors: 如 0x02010001：表示第一个电机超软限位
- status：program_run_finish/program_run_failed

## 8. get_config: 获取配置参数

Client

{

"command":"get_config",

}#

Server

{

"command": "get_config",

upperLimits" : [];

"lowerLimits": [];

status":"program_run_finish";

}#

- upperLimits: 软限位上限
- lowerLimits: 软限位下限
- status：program_run_finish/program_run_failed

### 9. 轨迹急停

Client

{

"command": "stop_motion"

}#

Server

{

"command": "stop_motion",

"status": "program_run_finish"

}#

说明：

" status":program_run_finish/program_run_failed

### 10. 清除错误

Client

{

"command": "clear_fault"

}#

Server

{

"command": "clear_fault",

"status": "program_run_finish"

}#

说明：

" status":program_run_finish/program_run_failed

### 11. 上使能

Client

{

"command": "enable_robot"

}#

Server

{

"command": "enable_robot",

"status": "program_run_finish"

}#

### 12. 下使能

Client

{

"command": "disable_robot"

}#

Server

{

"command": "disable_robot",

"status": "program_run_finish"

}#

说明：

" status":program_run_finish/program_run_failed

### 13. 下使能

Client

```
{
    "command": "disable_robot"
}#
```

Server

```
{
    "command": "disable_robot",
    "status": "program_run_finish"
}#
```

说明：

" status":program_run_finish/program_run_failed

### 14. 设置零位

Client

```
{
    "command": "set_zero"
}#
```

Server

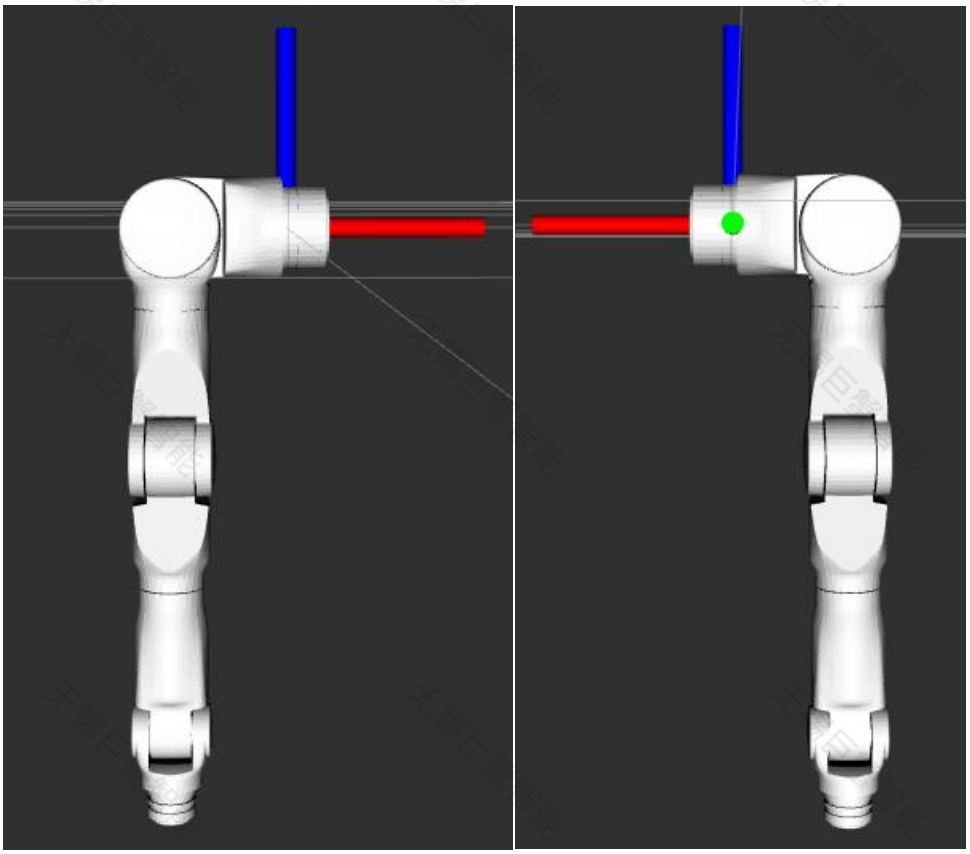
```
{
    "command": "set_zero",
    "status": "program_run_finish"
}#
```

说明：

" status":program_run_finish/program_run_failed

## 三、 坐标系说明

1. 基坐标为 base_link
2. tcp 坐标为 link7
3. 坐标系详见 urdf，示例



4. 快捷指令

- 7 轴

# movej

{"command":"movej","joint":[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],"v":80}#

{"command":"movej","joint":[0, 0.5, 0, 0, 1.1, 0, 0, 0, -0.5, 0, 0, 1.1, 0, 0, 0, 0, 0, 0],"v":80}#

# movej_p

{"command":"movej_p","pose":[-0.115764, 0.4426, -0.003899, -5.01103e-05, 1.21474e-05, 0.696694, 0.717368, -0.115696, -0.442622, -0.00385, 0.717406, 0.696655, -8.6642e-05, 0.000152392 ],"v":80}#

# movel
{"command":"movej","joint":[0, 0.5, 0, 0, 1.1, 0, 0, 0, -0.5, 0, 0, 1.1, 0, 0, 0, 0, 0],"v":80}#

{"command":"movel","pose":[-0.115764, 0.4426, -0.003899, -5.01103e-05, 1.21474e-05, 0.696694, 0.717368, -0.115696, -0.442622, -0.00385, 0.717406, 0.696655, -8.6642e-05, 0.000152392 ],"v":80}#

- 6 轴

# movej
{"command":"movej","joint":[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],"v":80}#
{"command":"movej","joint":[0, -0.5, 0, 0, 1.1, 0, 0, 0, -0.5, 0, 0, 1.1, 0, 0, 0, 0, 0],"v":80}#

# movej_p
{"command":"movej_p","pose":[0.009741, 0.110024, 0.385013, 0.697827, -7.92614e-05, -0.716266, -2.0242e-05, -0.115696, -0.442622, -0.00385, 0.717406, 0.696655, -8.6642e-05, 0.000152392],"v":80}#

# movel
{"command":"movej","joint":[0, -0.5, 0, 0, 1.1, 0, 0, 0, -0.5, 0, 0, 1.1, 0, 0, 0, 0, 0],"v":80}#
{"command":"movel","pose":[0.009741, 0.110024, 0.385013, 0.697827, -7.92614e-05, -0.716266, -2.0242e-05, -0.115696, -0.442622, -0.00385, 0.717406, 0.696655, -8.6642e-05, 0.000152392],"v":80}#

#enable_robot

{"command": "enable_robot"}#

#disenable_robot

{"command": "disenable_robot"}#

#stop
{"command":"stop_motion"}#

#clearFault
{"command": "clear_fault"}#

#get_joint_position

{"command": "get_joint_position"}#